Roll No.: AI.AI.U4AIR25100

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School of Artificial Intelligence, Faridabad
B.Tech Mid-Term Examinations – October 2025
First Semester
Artificial Intelligence & Data Science

# 23BIO112 Introduction to Biological Data

Maximum: 50 Marks

Duration: Two hours

**Course Outcomes (COs):**

| CO | Course Outcomes |
|---|---|
| CO01 | Develop ability to process (read and write) biological data. |
| CO02 | Analysis of different biological data |
| CO03 | Design AI and ML research problem to address biological research problem |
| CO04 | Develop knowledge to use different visualization tools and write script to handle software by command line |

**Answer all questions.**
**Draw the figure wherever it is needed while writing the answer.**

1 a) Write about DNA sequencing and protein sequencing. Why it is needful?
[5] [CO01] [BTL 2]

b) What are the FASTA and FASTQ? Discuss how these are useful in handling big biological data files?
[5] [CO01] [BTL 2]

2 What are the fundamental constituents for the development of organisms? Explain briefly with appropriate visuals.
[10] [CO02] [BTL 2]

3 What do you understand by biological data? Mention and brief about four different types of biological databases.
[10] [CO02] [BTL 2]

4 Write procedure for anemia detection of a patient using complete blood count (CBC).
[10] [CO02][BTL 3]



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neucleic acid

5 Write short notes on any four of the following:
(Each sub question carries 2.5 Marks). [10] [CO01] [BTL 2]

a. Genomics and proteomics
a. Base pairs or Nucleotides,
b. Drug development cycle,
c. Blood components,
d. Interpreting kidney function test

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**Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table**

| CO | Marks | BTL | Marks |
|---|---|---|---|
| CO01 | 20 | BTL 1 | - |
| CO02 | 30 | BTL 2 | 40 |
| CO03 | - | BTL 3 | 10 |
| CO04 | - | BTL 4 | - |



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